

FasterCap 2D Input Files

Dielectric definitions ('D' statement)

Syntax: D <file> <outperm> <inperm> <xoffset> <yoffset> <xref> <yref> [-]

The 'D' element at the beginning of a line defines a dielectric. The geometry of the dielectric contour is further specified in the file <file>. The dielectric contour is intended as the interface between two regions with relative permittivities <outperm> and <inperm>. The relative permittivities <outperm> and <inperm> can be complex valued, in the format $eRe - j eIm$, where eRe is the real part of the complex permittivity value, and eIm is the imaginary part. The dielectric can be translated with respect to the coordinates defined in <file> by an offset ($xoffset$, $yoffset$), thus allowing to reuse the same geometric definitions multiple times. The reference point ($xref$, $yref$) and the optional - argument are used to specify which side of the dielectric interface has which permittivity. More specifically, the reference point is assumed to lie on the <outperm> side of all the segments in <file>. The optional - argument indicates that ($xref$, $yref$) instead lies on the <inperm> side. The reference point is not translated, i.e. the offset only applies to the elements specified in <file>.

Remark: it is the user's responsibility to make sure that the reference point is on the same side of the dielectric interface for **all** panels. That is, each panel is evaluated as stand-alone with respect to the reference point, to define its <outperm> and <inperm> sides. There is no concept of a external and internal side of a surface specified by a group of panels, even if specified in the same file with the same conductor names, since FasterCap maintains no topological information.

Example 1:

```
D circle_0.15_fine.txt 1.0 2.0 0.0 0.0 0.0 0.0 -
```

This input file fragments specifies a dielectric interface whose geometry is defined in the file *circle_0.15_fine.txt*. The interior of the circle is filled with a material with relative permittivity equal to 2.0, since the reference point is centered on the origin and the optional - argument is specified; while the dielectric medium outside the circle has a relative permittivity equal to 1.0.

Example 2:

```
D circle_0.15_fine.txt 3.0-j0.02 1.0 0.0 0.0 0.0 0.0 -
```

This input file fragments specifies a dielectric interface whose geometry is defined in the file *circle_0.15_fine.txt*. The interface is between air (relative permittivity equal to 1.0) and a lossy dielectric with complex relative permittivity equal to $3.0 - j0.02$ (e.g. with real part equal to 3.0 and imaginary part equal to 0.02).